

Total No. of Questions : 10]

SEAT No. :

P2999

[Total No. of Pages : 3

[5669]-591

T.E. (Information Technology)
THEORY OF COMPUTATION
(2015 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8 and Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Define the following terms with an example. **[4]**

- i) Alphabet.
- ii) Regular Language.

b) Define formal definition of RE, also give the Regular expression for the following languages : **[6]**

- i) The Set a strings over the alphabet {a, b} starting with b and ending with odd number of a's or even numbers of b's.
- ii) The Set {10, 1010, ...}
- iii) If $L(r) = \{\epsilon, x, xx, xxx, xxxx, xxxxx\}$ what is r?

OR

Q2) a) Simplified the following grammar : **[4]**

$S \rightarrow Ab, A \rightarrow a, B \rightarrow C|b, C \rightarrow D, D \rightarrow E, E \rightarrow a$

- b) Discuss application of RE and FA. **[3]**
- c) Compare Moore machine and Mealy Machine. **[3]**

Q3) a) Construct the FA from given RE $1^*00(01)^*$ **[4]**

- b) Give CFG for following languages: **[6]**
 - i) Matching parenthesis
 - ii) All string without substring 'aaa'
 - iii) $R = bba * bb + bb$

P.T.O.

OR

Q4) a) Convert following left linear grammar to right linear grammar stepwise. [6]

$S \rightarrow A0 \mid B1$

$A \rightarrow C0 \mid A1 \mid 0$

$B \rightarrow B1 \mid A1 \mid 1$

$C \rightarrow 0 \mid A0$

b) State and prove that Pumping Lemma with an application. [4]

Q5) a) Design PDA to accept the language containing all odd length palindromes over $\Sigma = \{0,1\}$ by empty stack and final state. [8]

b) Design Post Machine that accept the following languages. [8]

$L = \{a^n b^n c^n \mid n \geq 0\}$

OR

Q6) a) Construct a PDA that accepts the language defined by the following grammar : [8]

$S \rightarrow 0A \mid 1B \mid 0$

$A \rightarrow 0A \mid B$

$B \rightarrow c \mid d$

Here $N = \{S, A, B\}$, $T = \{0, 1, c, d\}$ and S is start symbol.

b) Construct PDA by final state that accepts the following language. [8]

Simulate for "aaaaa" $L = \{a^{2n} \mid n > 0\}$

Q7) a) Design TM which compares two positive integers m & n and produces output [12]

Gt, if $m > n$; **Lt**, if $m < n$; and **Eq**, if $m = n$;

Simulate the working of the TM for the input i) $m = 1, n = 2$ ii) $m = n = 2$.

b) Write a note on each of the following : [6]

i) Recursively enumerable language.

ii) Recursive language

iii) Recursive function.

OR

Q8) a) Construct a TM to compute the following function. [12]

$f(a, b) = a - b$ where $a > b$

$= 0$ where $a < b$

Simulate the working of the TM for the input i) $x = 2, y = 2$. ii) $x = 4, y = 2$

b) Define TM. Explain its working. Give the types of TM & applications of the same. [6]

Q9) a) Suppose that there is an NP-complete problem P that has a deterministic solution taking $O(n^{\log 2^n})$ time (here $\log n$ denotes $\log 2^n$). What can you say about the running time of any other NP-complete problem Q? [9]

b) Prove that PCP with two lists $x = (b, bab^3, ba)$ and $y = (b^3, ba, a)$ have a solution? [7]

OR

Q10) a) Show that PCP over $|\Sigma| \geq 2$ is unsolvable. [9]

b) Find the running time for the Euclidean algorithm for evaluating GCD (a, b) where a, b are positive integers expressed in binary representation. [7]
